

Live Migration Demo

VMware → KVM via h2kweb Web Dashboard

Real migration results from a production Worldstream bare-metal server running AlmaLinux 9.7. All migrations performed through the h2kweb browser UI with PAM authentication, live log streaming, and automatic libvirt deployment.

✓ 3 VMs migrated & running — CentOS 10, Ubuntu 25.04, Ubuntu 24.04

April 1, 2026 — Worldstream NL (185.165.240.5)

Server Overview

Worldstream bare-metal server — production migration environment

8

CPU Cores

Intel Xeon E3-1230 v6 @ 3.50GHz

15.6 GB

Memory (47% used)

219 GB

Disk (94% used)

4

Libvirt Domains

System Configuration

Hostname	NLDW3-4-31-12
OS	AlmaLinux 9.7 (Moss Jungle Cat)
Kernel	5.14.0-611.41.1.el9_7.x86_64
QEMU	9.1.0
KVM	✓ Enabled
Uptime	3d 5h 20m
Load Average	4.59 / 4.53 / 4.44

Migrated VMs (all running)

centos10

CentOS Stream 10 (Coughlan) — LVM (cs-root/cs-home/cs-swap)

192.168.122.172

ubuntu-server2504

Ubuntu Server 25.04 (Plucky) — LVM (ubuntu-vg/ubuntu-lv)

192.168.122.136

ubuntu2404

Ubuntu 24.04 LTS — previously migrated

Migration: Ubuntu Server 25.04

VMDK → KVM with LVM, initramfs rebuild, auto-deploy to libvirt — 2m 21s total

VMDK

Source Format

QCOW2

Output Format

LVM

ubuntu-vg

2:21

Total Duration

YAML Configuration (auto-generated by h2kweb)

h2kweb → h2kvmctl1

```
command: local
vmdk: /root/demo/ubuntu-server2504.vmdk
output_dir: /var/lib/libvirt/images
out_format: qcow2
to_output: ubuntu-server2504.qcow2
flatten: true
compress: true
fstab_mode: stabilize-all
regen_initramfs: true
update_grub: true
remove_vmware_tools: true
emit_domain_xml: true
virsh_define: true
libvirt_test: true
keep_domain: true
vm_name: ubuntu-server2504
allowed_dirs:
  - /root/demo
  - /var/lib/libvirt/images
```

Pipeline Timeline

Phase	Duration	Details
Flatten (VMDK → QCOW2)	56s	qemu-img convert, monolithicSparse
NBD Connect	0.7s	/dev/nbd0, qcow2 format
LVM Activation	0.7s	ubuntu-vg: 1 LV (ubuntu-lv)
OS Detection	0.3s	Ubuntu 25.04 (Plucky) on ext4
fstab Stabilize	<0.1s	UUID-based (already stable)
Network Fix	<0.1s	NM connection fixed
initramfs Rebuild	11s	update-initramfs -u -k all
GRUB Update	3s	grub-mkconfig + grub-install

Final Convert	51s	Compressed QCOW2 output
Libvirt Deploy	1s	virsh define + start → RUNNING

Result



ubuntu-server2504

Defined, started, and running in libvirt

● **192.168.122.136**

ssh root@192.168.122.136

Migration: CentOS Stream 10

LVM-based CentOS 10 with dracut initramfs rebuild and virtio driver injection

VMDK

Source

1.24 GB

Compressed QCOW2

3 LVs

cs-root / cs-home / cs-swap

1:45

Total Duration

Key Pipeline Events

h2kweb Live Output – 380 lines

```

[✓] Quick probe: detected LVM on disk
[✓] Container scan found 1 VG(s): cs
[✓] Found 3 LVs: cs-root, cs-home, cs-swap
[✓] Detected: CentOS Stream 10 (Coughlan)
[✓] Mounted root: /dev/mapper/cs-root (xfs, score=43)
[✓] fstab: 4 entries, all UUID-stable
[✓] Network: ens33.nmconnection fixed (2 fixes)
[✓] Masked: vmtoolsd.service, vgauthd.service
🔧 dracut -f --kver 6.12.0-43.el10.x86_64
  --add-drivers virtio_blk virtio_scsi virtio_net
  --add lvm dm
[✓] Completed in 26.3s
[✓] grub2-mkconfig: Succeeded
[✓] Blacklisted VMware kernel modules
[✓] Conversion complete: centos10.qcow2
[✓] Domain centos10 → RUNNING

```

Automated Fixes Applied

Fix	Status
LVM detection (cs VG)	✓ 3 LVs activated
Root FS detection	✓ /dev/mapper/cs-root (xfs)
fstab stabilization	✓ UUID-based (4 entries)
Network fix	✓ ens33 NM connection
VMware tools removal	✓ vmtoolsd, vgauthd masked
fstab hardening	✓ /boot, /home → nofail
initramfs rebuild (dracut)	✓ 26.3s with virtio + lvm
GRUB regeneration	✓ grub2-mkconfig
LVM filter fix	✓ Removed restrictive filter

SELinux autorelabel ✓ .autorelabel created

VMware module blacklist ✓ Kernel modules blocked

Serial console ✓ Injected in default/grub

Auto-grow root FS ✓ Configured (xfs)

Machine-ID reset ✓ Firstboot regeneration



centos10

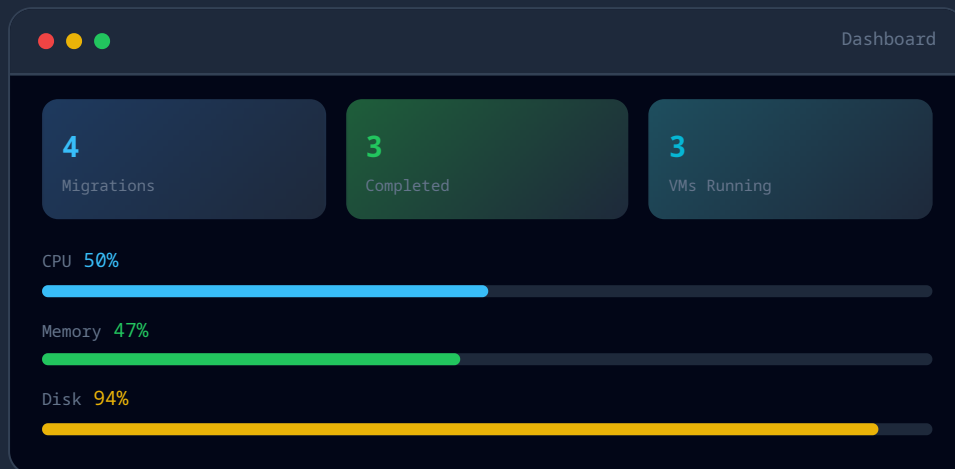
CentOS Stream 10 — LVM + virtio + BIOS

● 192.168.122.172

Web Dashboard — h2kweb

Browser-based migration management at <http://185.165.240.5:5070>

Dashboard View



Migration Wizard

Migrate - Local File

Local vSphere Azure EC2

/ root / demo

- centos10.vmdk VMDK 2.2 GB
- centos9.vmdk VMDK 1.8 GB
- ubuntu-server2504.vmdk VMDK ✓

Next: Configure →

Job Monitor — Live Logs

Job c690a469 • streaming

- ✓ Detected LVM on disk
- ✓ Ubuntu 25.04 (Plucky)
- 🔧 update-initramfs -u -k all
- ✓ grub-mkconfig: Succeeded
- ✓ Conversion complete
- ✓ Domain → RUNNING

192.168.122.136 SSH

Providers Page

Providers

- VMware vSphere**
govmomi SDK
▶ Sign in to connect
- Microsoft Azure**
az CLI
▶ Sign in to connect
- AWS EC2**
aws CLI
▶ Sign in to connect

Summary & Capabilities

Production-validated VM migration with full automation

What Was Demonstrated

Capability	Result
VMware VMDK → KVM QCOW2	✓ Automated
LVM detection & activation	✓ cs, ubuntu-vg
Root filesystem detection	✓ xfs, ext4
initramfs rebuild (dracut/update-initramfs)	✓ virtio drivers
GRUB bootloader repair	✓ grub2-mkconfig
fstab stabilization (UUID)	✓ All entries
Network config fix	✓ NM connections
VMware tools removal	✓ Services masked
Auto-deploy to libvirt	✓ virsh define+start
VM boots with IP	✓ DHCP assigned
Web UI (PAM auth)	✓ h2kweb :5070
Live log streaming	✓ WebSocket

Guest OS Support Validated

CentOS 10
Stream • LVM • dracut

Ubuntu 25.04
Server • LVM • update-initramfs

Ubuntu 24.04
LTS • ext4 • GRUB

CentOS 9
Stream • xfs • dracut

Deployment

One-command deploy

```
$ ./scripts/deploy-remote.sh 185.165.240.5 root

# Installs:
h2kvmctl (Python CLI)
h2kweb (Go web dashboard)
systemd services
libguestfs auto-linking
augeas for config editing

✓ Dashboard: http://185.165.240.5:5070
✓ Login with system PAM credentials
```

